Naïve Bayes Classifier

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Naïve Bayes Classifier

- Machine Learning
  - Supervised Learning
    - Classification
  - Global Learning
  - Model-Based Learning
  - Eager Learning

Naïve Bayes Classifier

Bayes' Theorem?

Probability

- A **probability** is a number that represents the likelihood of an uncertain event.
- Probabilities are always between 0 and 1, inclusive.
  - The larger the probability, the more likely the event is to happen.
  - A probability of 0 means that the event is impossible; it will never happen.
  - A probability of 1 means that the event is assured; it will always happen.
  - All other values between 0 and 1 represent various levels of likelihood.
- Probability of occurrence of an event ***P(E)***:

$$P(E) = \frac{Number\ of\ favorable\ outcomes}{Total\ Number\ of\ outcomes}$$

General Probability Rules *(1-2 of 6)*

- **Rule 1:**
  - The probability of an impossible event is zero;
  - the probability of a certain event is one.
  - Therefore, for any event A, the range of possible probabilities is:

$$0 \leq P(A) \leq 1$$

- **Rule 2:**
  - For S the sample space of all possibilities, P(S) = 1.
  - That is the sum of all the probabilities for all possible events is equal to one.
  - Recall the party affiliation above: if you have to belong to one of the three designated political parties, then the sum of P(R), P(D) and P(I) is equal to one.

General Probability Rules *(3-4 of 6)*

- **Rule 3:**
  - For any event *A*, $P(A') = 1 - P(A)$.
  - It follows then that $P(A) = 1 - P(A')$.
- **Rule 4 (Addition Rule):**
  - This is the probability that ***either one or both*** events occur

a) If two events, say *A* and *B*, are ***mutually exclusive*** - that is *A* and *B* have no outcomes in common – then:

$$P(A \cup B) = P(A) + P(B)$$

a) If two events are ***NOT mutually exclusive***, then:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

General Probability Rules *(5-6 of 6)*

- **Rule 5 (Multiplication Rule):**
  - This is the probability that ***both*** events occur

  a) $P(A \cap B) = P(A)P(B|A)$ and

  b) $P(B \cap A) = P(B)P(A|B)$ and
  $P(A \cap B) = P(B \cap A)$

  Note: this straight line symbol, **|**, does not mean divide!

  This symbols means "*conditional*" or "*given*".

  For instance $P(A|B)$ means the probability that event *A* occurs given event *B* has occurred.

  c) If *A* and *B* are independent - neither event influences or affects the probability that the other event occurs - then $P(A \cap B) = P(A)P(B)$.

  This particular rule extends to more than two independent events.

  For example, $P(A \cap B \cap C) = P(A)P(B)P(C)$

- **Rule 6 (Conditional Probability):**

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \text{ or } P(B|A) = \frac{P(B \cap A)}{P(A)}$$

Bayes' Theorem

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, if\ P(B) \neq 0,$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}, if\ P(A) \neq 0,$$

$$P(A \cap B) = P(B \cap A)$$

$$\Rightarrow P(A|B)P(B) = P(A \cap B) = P(B \cap A) = P(B|A)P(A)$$

$$\Rightarrow P(A|B) = \frac{P(B|A)P(A)}{P(B)}, if\ P(B) \neq 0.$$

Bayes' Theorem

$$P(A|B) = \frac{\boldsymbol{P(A \cap B)}}{P(B)}, if\ P(B) \neq 0,$$

$$P(B|A) = \frac{P(B \cap A)}{P(A)}, if\ P(A) \neq 0,$$

$$\boldsymbol{P(A \cap B) = P(B \cap A)}$$

$$\Rightarrow P(A|B)P(B) = \boldsymbol{P(A \cap B) = P(B \cap A) = P(B|A)P(A)}$$

$$\Rightarrow P(A|B) = \frac{\boldsymbol{P(B|A)P(A)}}{P(B)}, if\ P(B) \neq 0.$$

Bayes' Theorem

$$P(A|B) = \frac{P(B|A)P(A)}{P(B)}$$

Probabilistic Model

- Abstractly, ***Naïve Bayes*** is a conditional probability model: given a problem instance to be classified, represented by a vector ***x = $(x_1, \ldots, x_n)$*** representing some ***n*** features (independent variables), it assigns to this instance probabilities

$$p(C_k | x_1, \ldots, x_n)$$

for each of **K** possible outcomes or *classes* $\boldsymbol{C_K}$.

Probabilistic Model

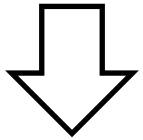
- The problem with the above formulation is that if the number of features ***n*** is large or if a feature can take on a large number of values, then basing such a model on probability tables is infeasible. We therefore reformulate the model to make it more tractable. Using Bayes' theorem, the conditional probability can be decomposed as

$$p(C_k|x) = \frac{p(C_k)p(x|C_k)}{p(x)}$$

Probabilistic Model

- In plain English, using Bayesian probability terminology, the above equation can be written as

$$\boldsymbol{posterior} = \frac{\boldsymbol{prior} \times \boldsymbol{likelihood}}{\boldsymbol{evidence}}$$



$$P(Hypothesis|Data) = \frac{P(Hypothesis) \times P(Data|Hypothesis)}{P(Data)}$$

Probabilistic Model

- In practice, there is interest only in the numerator of that fraction, because the denominator does not depend on ***C*** and the values of the features $\boldsymbol{x_i}$ are given, so that the denominator is effectively constant. The numerator is equivalent to the ***joint probability*** model

$$p(C_k, x_1, \ldots, x_n)$$

Probabilistic Model

- which can be rewritten as follows, using the ***chain rule*** for repeated applications of the definition of ***conditional probability***:

$$
\begin{aligned}
p(C_k, x_1, \ldots, x_n) &= p(x_1, \ldots, x_n, C_k) \\
&= p(x_1 \mid x_2, \ldots, x_n, C_k)\, p(x_2, \ldots, x_n, C_k) \\
&= p(x_1 \mid x_2, \ldots, x_n, C_k)\, p(x_2 \mid x_3, \ldots, x_n, C_k)\, p(x_3, \ldots, x_n, C_k) \\
&= \cdots \\
&= p(x_1 \mid x_2, \ldots, x_n, C_k)\, p(x_2 \mid x_3, \ldots, x_n, C_k) \cdots p(x_{n-1} \mid x_n, C_k)\, p(x_n \mid C_k)\, p(C_k)
\end{aligned}
$$

Probabilistic Model

- Now the "naïve" conditional independence assumptions come into play: assume that all features in ***x*** are mutually independent, conditional on the category $\boldsymbol{C_k}$. Under this assumption,

$$p(x_i|x_{i+1}, \dots, x_n, C_k) = p(x_i|C_k)$$

Probabilistic Model

- Thus, the joint model can be expressed as

$$
\begin{aligned}
p(C_k \mid x_1, \ldots, x_n) &\propto p(C_k, x_1, \ldots, x_n) \\
&\propto p(C_k)\, p(x_1 \mid C_k)\, p(x_2 \mid C_k)\, p(x_3 \mid C_k)\, \cdots \\
&\propto p(C_k) \prod_{i=1}^{n} p(x_i \mid C_k)\,,
\end{aligned}
$$

where $\propto$ denotes proportionality.

Probabilistic Model

- This means that under the above independence assumptions, the conditional distribution over the class variable ***C*** is:

$$p(C_k \mid x_1, \ldots, x_n) = \frac{1}{Z} p(C_k) \prod_{i=1}^{n} p(x_i \mid C_k)$$

where the evidence $Z = p(\mathbf{x}) = \sum_k p(C_k)\, p(\mathbf{x} \mid C_k)$

is a scaling factor dependent only on $x_1,\ldots,x_n$, that is, a constant if the values of the feature variables are known.

Constructing a Classifier from the Probability Model

The discussion so far has derived the independent feature model, that is, the naïve Bayes probability model. The naïve Bayes classifier combines this model with a decision rule. One common rule is to pick the hypothesis that is most probable; this is known as the maximum a posteriori or MAP decision rule. The corresponding classifier, a Bayes classifier, is the function that assigns a class label $\hat{y} = C_k$ for some k as follows:

$$\hat{y} = \underset{k \in \{1,\dots,K\}}{\operatorname{argmax}} \; p(C_k) \prod_{i=1}^{n} p(x_i \mid C_k)$$

Example

http://shatterline.com/blog/2013/09/12/not-so-naive-classification-with-the-naive-bayes-classifier/

The Dataset

	Predictors				Response
	Outlook	**Temperature**	**Humidity**	**Wind**	**Class** **Play=Yes** **Play=No**
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

The Dataset

- **Outlook** ∈ [Sunny, Overcast, Rainy]
- **Temperature** ∈ [Hot, Mild, Cool]
- **Humidity** ∈ [High, Normal]
- **Windy** ∈ [Weak, Strong]

- **Play** ∈ [Yes, No]

The Probabilities We Need

- **The Prior:** $P(C_k)$
- **The Likelihood:** $P(x_i|C_k)$

The Probabilities We Need

- **The Prior:** $P(C_k)$
  - **P(Class$_{\text{Play=Yes}}$)** and **P(Class$_{\text{Play=No}}$)**

- **The Likelihood:** $P(x_i|C_k)$
  - **P(Outlook=o|Class$_{\text{Play=b}}$)**
    where **o ∈ [Sunny, Overcast, Rainy]** and **b ∈ [yes, no]**
  - **P(Temperature=t|Class$_{\text{Play=b}}$)**
    where **t ∈ [Hot, Mild, Cool]** and **b ∈ [yes, no]**
  - **P(Humidity=h|Class$_{\text{Play=b}}$)**
    where **h∈ [High, Norma]** and **b ∈ [yes, no]**
  - **P(Wind=w|Class$_{\text{Play=b}}$)**
    where **w ∈ [Weak, Strong]** and **b ∈ [yes, no]**

Naïve Bayes Model

P(Outlook=o|Class $_{Play=Yes\|No}$)	Frequency		Probability in Class	
Outlook =	Play=Yes	Play=No	Play=Yes	Play=No
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5
	total= 9	total=5		

P(Temperature=t|Class $_{Play=Yes\|No}$)	Frequency		Probability in Class	
Temperature =	Play=Yes	Play=No	Play=Yes	Play=No
Hot	2	2	2/9	2/5
Mild	4	2	4/9	2/5
Cool	3	1	3/9	1/5
	total= 9	total=5		

P(Humidity=h|Class $_{Play=Yes\|No}$)	Frequency		Probability in Class	
Humidity =	Play=Yes	Play=No	Play=Yes	Play=No
High	3	4	3/9	4/5
Normal	6	1	6/9	1/5
	total= 9	total=5		

P(Wind=w|Class $_{Play=Yes\|No}$)	Frequency		Probability in Class	
Wind =	Play=Yes	Play=No	Play=Yes	Play=No
strong	3	3	3/9	3/5
weak	6	2	6/9	2/5
	total= 9	total=5		

P(Class $_{Play=Yes}$) & P(Class $_{Play=No}$)	Frequency		Probability in Class	
Play	Play=Yes	Play=No	Play=Yes	Play=No
	9	5	9/14	5/14
	total= 9	total=5		

The Test Sample "$\boldsymbol{x_t}$"

- $\boldsymbol{x_t}$ = (**Outlook** = *Sunny*,
  Temperature = *Cool*,
  Humidity = *High*,
  Wind = *Strong*)

What Exactly We Calculate

- $P(Class_{Play=Yes}|x_t) = [P(Sunny|Class_{Play=Yes})$
  $\times P(Cool|Class_{Play=Yes})$
  $\times P(High|Class_{Play=Yes})$
  $\times P(Strong|Class_{Play=Yes})]$
  $\times P(Class_{Play=Yes})$
  $= 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14$
  $= 0.0053$

- $P(Class_{Play=No}|x_t) = [P(Sunny|Class_{Play=No})$
  $\times P(Cool|Class_{Play=No})$
  $\times P(High|Class_{Play=No})$
  $\times P(Strong|Class_{Play=No})]$
  $\times P(Class_{Play=No})$
  $= 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14$
  $= 0.0205$

How Exactly We Do It *(Class)*

- **The Prior:** $P(C_k)$
  - **$P(Class_{Play=Yes})$** and **$P(Class_{Play=No})$**

P(Class $_{Play=Yes}$) & P(Class $_{Play=No}$)	Frequency		Probability in Class	
Play	Play=Yes	Play=No	Play=Yes	Play=No
	9	5	9/14	5/14
	total= 9	total=5		

	Predictors				Response
	Outlook	**Temperature**	**Humidity**	**Wind**	**Class Play=Yes Play=No**
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
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Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

$$P(Class_{Play=Yes} | x_t) = [P(Sunny | Class_{Play=Yes}) \times P(Cool | Class_{Play=Yes}) \times P(High | Class_{Play=Yes}) \times P(Strong | Class_{Play=Yes})] \times P(Class_{Play=Yes})$$
$$= 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14$$
$$\mathbf{= 0.0053}$$

$$P(Class_{Play=No} | x_t) = [P(Sunny | Class_{Play=No}) \times P(Cool | Class_{Play=No}) \times P(High | Class_{Play=No}) \times P(Strong | Class_{Play=No})] \times P(Class_{Play=No})$$
$$= 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14$$
$$\mathbf{= 0.0205}$$

How Exactly We Do It *(Outlook)*

- **The Likelihood:** $P(x_i|C_k)$
  - **P(Sunny|Class$_{Play=Yes}$)** and **P(Sunny|Class$_{Play=No}$)**

P(Outlook=o|Class $_{Play=Yes\|No}$)	Frequency		Probability in Class	
Outlook =	Play=Yes	Play=No	Play=Yes	Play=No
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5
	total= 9	total=5		

	Predictors				Response
	Outlook	Temperature	Humidity	Wind	Class Play=Yes Play=No
Day1	Sunny	Hot	High	Weak	No
Day2	Sunny	Hot	High	Strong	No
Day3	Overcast	Hot	High	Weak	Yes
Day4	Rain	Mild	High	Weak	Yes
Day5	Rain	Cool	Normal	Weak	Yes
Day6	Rain	Cool	Normal	Strong	No
Day7	Overcast	Cool	Normal	Strong	Yes
Day8	Sunny	Mild	High	Weak	No
Day9	Sunny	Cool	Normal	Weak	Yes
Day10	Rain	Mild	Normal	Weak	Yes
Day11	Sunny	Mild	Normal	Strong	Yes
Day12	Overcast	Mild	High	Strong	Yes
Day13	Overcast	Hot	Normal	Weak	Yes
Day14	Rain	Mild	High	Strong	No

$P(Class_{Play=Yes}|x_t) = [P(Sunny|Class_{Play=Yes})$
$\times P(Cool|Class_{Play=Yes})$
$\times P(High|Class_{Play=Yes})$
$\times P(Strong|Class_{Play=Yes})]$
$\times P(Class_{Play=Yes})$
$= 2/9 \times 3/9 \times 3/9 \times 3/9 \times 9/14$
= 0.0053

$P(Class_{Play=No}|x_t) = [P(Sunny|Class_{Play=No})$
$\times P(Cool|Class_{Play=No})$
$\times P(High|Class_{Play=No})$
$\times P(Strong|Class_{Play=No})]$
$\times P(Class_{Play=No})$
$= 3/5 \times 1/5 \times 4/5 \times 3/5 \times 5/14$
= 0.0205

How Exactly We Classify

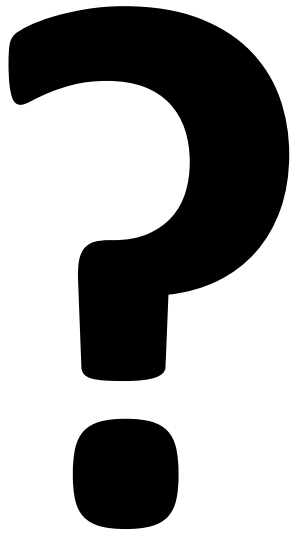
- $\mathbf{P(Class_{Play=Yes} | x_t)}$ = [P(Sunny|$Class_{Play=Yes}$)
  × P(Cool|$Class_{Play=Yes}$)
  × P(High|$Class_{Play=Yes}$)
  × P(Strong|$Class_{Play=Yes}$)]
  × P($Class_{Play=Yes}$)
  = 2/9 × 3/9 × 3/9 × 3/9 × 9/14
  = 0.0053

- $\mathbf{P(Class_{Play=No} | x_t)}$ = [P(Sunny|$Class_{Play=No}$)
  × P(Cool|$Class_{Play=No}$)
  × P(High|$Class_{Play=No}$)
  × P(Strong|$Class_{Play=No}$)]
  × P($Class_{Play=No}$)
  = 3/5 × 1/5 × 4/5 × 3/5 × 5/14
  = 0.0205

- $\mathbf{P(Class_{Play=Yes} | x_t) \leq P(Class_{Play=No} | x_t)}$

  ⇒ we classify the new instance $\mathbf{x_t}$ to be "**No**"

Zero Probability Problem



Zero Probability Problem I

P(Outlook=o | Class $_{\text{Play=Yes \| No}}$)	Frequency		Probability in Class	
Outlook =	Play=Yes	Play=No	Play=Yes	Play=No
Sunny	2	3	2/9	3/5
Overcast	4	0	4/9	0/5
Rain	3	2	3/9	2/5
	total= 9	total=5		

Laplace Smoothing

In statistics, **Laplace Smoothing** is a technique to smooth categorical data. Laplace Smoothing is introduced to solve the problem of zero probability. By applying this method, prior probability and conditional probability can be written as:

$$p_\lambda(C_k) = p_\lambda(Y = C_k) = \frac{\sum_{i=1}^{N} I(y_i = C_k) + \lambda}{N + K\lambda}$$

$$p(x_1 = a_j | y = C_k) = \frac{\sum_{i=1}^{N} I(x_{1i} = a_j, y_i = C_k) + \lambda}{\sum_{i=1}^{N} I(y_i = C_k) + A\lambda}$$

K denotes the number of different values in ***y*** and ***A*** denotes the number of different values in $\boldsymbol{a_j}$. Usually, ***lambda*** in the formula equals to 1.

Zero Probability Problem II

- 0.1
- $0.1 \times 0.2 = 0.02$
- $0.02 \times 0.3 = 0.006$
- $0.006 \times 0.4 = 0.0024$
- $0.0024 \times 0.5 = 0.0012$
- $0.0012 \times 0.6 = 0.00072$

- $0.00072 \times 0.1 = 0.000072$

What about ***N*** *features?*

Log Probabilities

$$\log(ab) = \log(a) + \log(b)$$

$$\log(P(\text{class }_i|\ \textbf{data})) \propto \log(P(\text{class}_i)) + \sum_j \log(P(\text{data}_j|\text{class}_i))$$

Gaussian Naïve Bayes

$$p(x = v|C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(v-\mu_k)^2}{2\sigma_k^2}}$$

Gaussian Naïve Bayes

$$p(x = v|C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(v-\mu_k)^2}{2\sigma_k^2}}$$

Gaussian Naïve Bayes

	sex	height	weight	foot size							
	male	6.00	180	12							
	male	5.92	190	11							
	male	5.58	170	12							
	male	5.92	165	10							
AVG		**5.86**	**176.25**	**11.25**							
STDEV		**0.19**	**11.08677891**	**0.957427108**			**Product**				
NORM.DIST		**0.11**	**0.000171654**	**0.177691643**			1.69339E-06				
		5.40	**140**	**10**	**0.5**						
									MAX	**0.000485934**	**female**
	sex	**height**	**weight**	**foot size**							
	female	5.00	100	6							
	female	5.50	150	8							
	female	5.42	130	7							
	female	5.75	150	9							
AVG		**5.42**	**132.5**	**7.5**							
STDEV		**0.311809237**	**23.62907813**	**1.290994449**			**Product**				
NORM.DIST		**1.277429954**	**0.01605412**	**0.047389654**			0.000485934				

$$p(x = v|C_k) = \frac{1}{\sqrt{2\pi\sigma_k^2}} e^{-\frac{(v-\mu_k)^2}{2\sigma_k^2}}$$

Naïve Bayes Classifier Implementations *(scikit-learn)*

- Gaussian Naive Bayes
  - **`GaussianNB`**
- Multinomial Naive Bayes
  - **`MultinomialNB`**
- Complement Naive Bayes
  - **`ComplementNB`**
- Bernoulli Naive Bayes
  - **`BernoulliNB`**
- Categorical Naive Bayes
  - **`CategoricalNB`**
- Out-of-core naive Bayes model fitting
  - **`MultinomialNB`**, **`BernoulliNB`**, and **`GaussianNB`** expose a **`partial_fit`** method that can be used incrementally.

* *https://scikit-learn.org/stable/modules/naive_bayes.html*

Homework: NBC

Implement a Naive Bayesian Classifier that classifies individuals as Democrats or Republicans using the 16 attributes and the two classes of the following data:

- http://archive.ics.uci.edu/dataset/105/congressional+voting+records

Addition to the task:

- To test the algorithm, apply 10-fold cross-validation (divide the data randomly into 10 sets and do 10 trainings and tests of the model, each time using one of the sets for testing and the other 9 for training).
- Display the Accuracy metric for each of the 10 trainings, as well as their arithmetic mean (for a summary assessment of the classifier).

** Data may need pre-processing.*